Olympus: Risk-Monotone Counterfactual Fusion for Governed Multimodal Prediction

Registered Architecture and Bounded Engineering Evidence

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Abstract

Specialized predictors can expose complementary linguistic, scientific, multimodal, temporal, and governance signals, yet learned fusion may hide routing collapse and allow model confidence to weaken external safety information. We specify Olympus, a reference architecture that projects heterogeneous component states into a common space, emits leave-one-component-out decisions, derives an intrinsic risk estimate from representation disagreement, and combines that estimate with a host risk floor through a monotone noisy-OR. Host-supplied legal-transition masks and a deterministic STOP threshold remain outside learned authority. We also specify a geometry-conditioned joint-embedding predictive super-resolution subsystem. The central scientific hypotheses were not tested: the repository lacks a rights-reviewed aligned corpus and its small smoke test was already observed. We instead report a preregistered 24-run engineering study on 36 proprietary instruction records. Cosine scheduling changed observed test loss by -0.00969 relative to a constant rate (paired bootstrap 95% interval $[-0.01607, -0.00564]$), while restoring the validation-best state changed it by -0.02784 relative to the final epoch ($[-0.03644, -0.02087]$). Exact match was 0% in every run. The scientific disposition is therefore inconclusive for Olympus and mixed for bounded engineering. The main output is a falsifiable model and evaluation contract, an implementation, and a complete negative and blocked-evidence record rather than a capability claim.

1 Introduction

Multimodal systems are often evaluated under complete, clean observations even though missingness, corruption, imbalance, and sample-dependent quality are normal in deployed settings. Empirical work shows that multimodal Transformers can be sensitive to missing modalities and that the best fusion strategy is dataset-dependent [8, 9]. Mixture-of-experts systems introduce another failure surface: gates can collapse, experts can be under-trained, and balancing losses can interfere with the task objective [4, 11, 10]. Uncertainty and selective prediction offer useful diagnostics, but neither ordinary confidence thresholds nor an architectural score is automatically a calibrated safety guarantee [5, 1, 6].

We study the following question: can heterogeneous predictors be fused such that marginal component influence is observable, missing-component sensitivity is directly measurable, and learned evidence may increase but never decrease a host-declared risk score? The precise research statement is:

> Existing fusion approaches struggle to remain auditable and robust when heterogeneous components are missing, corrupted, or safety-relevant. We propose host-monotone counterfactual fusion because explicit component interventions and monotone risk composition predict improved failure detection without allowing learned evidence to weaken external constraints.

The candidate novelty is the combined contract, not any ingredient alone. This paper separates the registered scientific architecture from a bounded instruction-model case study. The latter tests training infrastructure only.

2 Related work and novelty boundary

Joint-embedding predictive architectures learn by predicting target representations from context rather than reconstructing every pixel. I-JEPA established this principle for images [2]; V-JEPA 2 extended it to video, action-

conditioned prediction, and planning [3]. The Olympus upscaler borrows the predictive-representation principle but is not a reproduction of either model.

Sparse MoE work studies conditional computation and routing stability. Switch uses simple top-1 routing, Expert Choice lets experts select tokens under fixed capacity, and loss-free balancing adjusts expert-wise routing bias without auxiliary-loss gradients [4, 11, 10]. Olympus presently uses dense sample-level soft gating, so it cannot claim sparse-compute efficiency.

SelectiveNet jointly learns prediction and rejection and evaluates risk-coverage tradeoffs [5]. Deep ensembles provide a strong uncertainty and shift baseline [6]. Conformal Risk Control controls the expectation of a monotone loss under its calibration assumptions [1]. Olympus's monotone noisy-OR is only an algebraic invariant; it does not provide conformal coverage.

Our search found no exact prior system combining normalized heterogeneous fusion, first-class leave-one-component-out predictions, disagreement-conditioned risk, host-monotone risk composition, and host-owned legal transitions. This is a provisional non-collision result. Independent literature verification remains a release gate.

3 Problem formulation

For one independent unit x, let m bounded component encoders emit $h_i \in \mathbb{R}^{d_i}$. Components may represent text, scientific evidence, workflow traces, images, temporal state-action histories, or registered extension channels. The system predicts a task label $y \in \{1, \dots, C\}$, a next temporal state, a legal research transition, intrinsic failure risk, and abstention/STOP availability.

The host supplies risk $r_h \in [0, 1]$ and a nonempty Boolean legal-transition set $A(x)$. These are trusted inputs, not model outputs. Training assumes independent source units, finite tensors, valid labels, and source-disjoint train, calibration, validation, and test sets. Identifiability of any component's causal contribution is not assumed; leave-one-out effects are diagnostics unless verified by executed interventions.

The scientific objective is to minimize task, transition, temporal, uncertainty, and risk losses while controlling routing collapse and component-removal sensitivity. The primary confirmatory estimand is the paired difference between Olympus and the strongest matched baseline on a frozen task metric. Secondary estimands cover selective risk, calibration, corruption robustness, compute, memory, and failure rate.

4 Model specification

4.1 Bounded component families

Hermes maps byte/token sequences to instruction and tool representations. Prometheus adds domain adapters and evidence classes. Perseus uses a routed expert bank for workflow states. Atlas combines token and image-patch representations with retrieval coordinates. Kronos maps ordered state-action sequences to next-state predictions. Aion maps current state, effective risk, and host legal transitions to a masked next-stage distribution. Each reference family is independently executable and shape-checked. Family names describe interfaces, not frontier capability.

4.2 Normalized gated fusion

Each component is projected into a shared width d and normalized:

$$z_i = \mathrm{LayerNorm}(W_i h_i + b_i), \tag{1}$$

$$g = \mathrm{softmax}\left(r([z_1; \dots; z_m])/\tau\right), \tag{2}$$

$$z = \mathrm{LayerNorm}\left(\sum_{i=1}^{m} g_i z_i\right), \tag{3}$$

where $\tau > 0$ is fixed before confirmation. The implementation emits g, every z_i, and the fused z.

4.3 Counterfactual component output

For $\epsilon > 0$, the leave-one-component-out state is

$$z_{-i} = \mathrm{LayerNorm}\left(\frac{\sum_j g_j z_j - g_i z_i}{\max(1 - g_i, \epsilon)}\right). \tag{4}$$

The shared decision head evaluates both z and every z_{-i}. A training regularizer may use

$$L_{\mathrm{cf}} = \mathbb{E}_b \sum_i \mathrm{stopgrad}(g_{bi}) D_{\mathrm{KL}}\left(\mathrm{stopgrad}(p_b) \,\|\, p_{b,-i}\right). \tag{5}$$

Because the full prediction is stop-gradient, this term encourages removal robustness; it does not prove specialization or causal necessity.

4.4 Disagreement and host-monotone risk

Let c be the normalized gated centroid. Disagreement and intrinsic risk are

$$d = \frac{1}{2} \sum_i g_i \left[1 - \cos(z_i, c)\right], \quad 0 \le d \le 1, \tag{6}$$

$$q = \sigma(f([z; d])). \tag{7}$$

With host risk r_h, effective risk is

$$r_{\mathrm{eff}} = 1 - (1 - r_h)(1 - q). \tag{8}$$

Since $r_{\mathrm{eff}} - r_h = q(1 - r_h) \ge 0$, learned state cannot lower the host risk floor. This proof is pointwise for bounded inputs. Calibration requires independent labels and a separate calibration procedure.

4.5 Governed transitions

Aion assigns negative infinity to transitions outside $A(x)$. If r_{eff} reaches the fixed kill threshold, the selected transition is STOP if STOP is host-allowed; otherwise execution raises an error. Forced STOP zeroes the externally exposed decision logits. Raw logits remain available for audit and loss computation but cannot authorize execution.

4.6 Objective and complexity

The joint objective is

$$L = L_{\mathrm{decision}} + \lambda_t L_{\mathrm{transition}} + \lambda_k L_{\mathrm{temporal}} + \lambda_u L_{\mathrm{hetero}} + \lambda_r L_{\mathrm{risk}} + \lambda_b D_{\mathrm{KL}}(\bar{g} \| U) + \lambda_{\mathrm{cf}} L_{\mathrm{cf}}. \tag{9}$$

All weights are finite and nonnegative. The fusion projection costs $O(\sum_i d_i d)$; dense gating and fusion cost $O(md)$ per example; producing all counterfactual logits costs $O(mCd)$ in the reference implementation. Component backbone costs dominate at target scale.

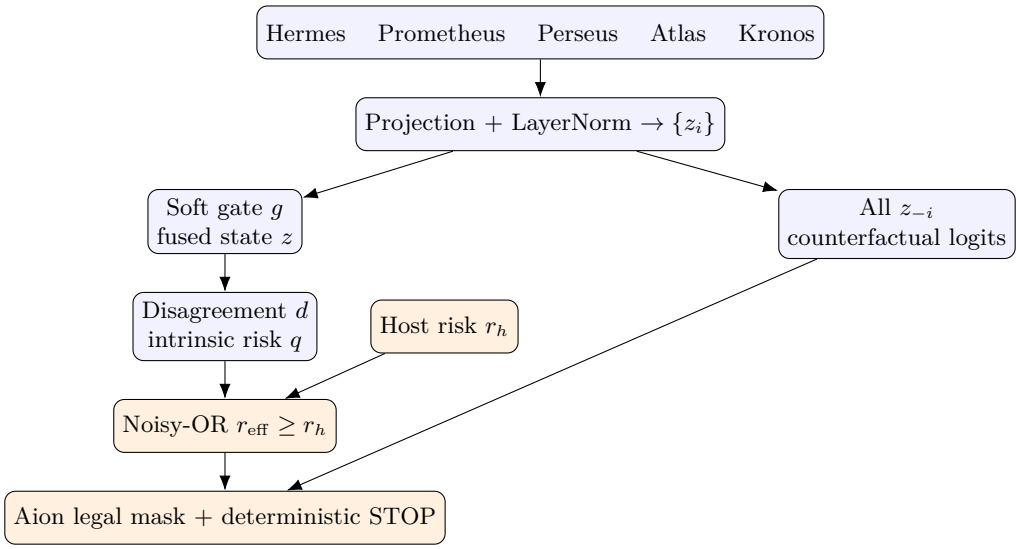


Figure 1: Olympus separates learned fusion and intrinsic risk from host-owned risk and transition authority. Every component-removal output remains observable.

5 JEPA super-resolution subsystem

The upscaler predicts exponential-moving-average target embeddings for high-resolution patches from masked bicubic context and explicit spatial/temporal coordinates. Its registered objective combines Charbonnier pixel error, edge error, normalized latent prediction, latent-gradient geometry, a variance floor, log-magnitude spectrum error, SSIM, multiscale pyramid consistency, and target-magnitude-weighted phase coherence. Tiled inference bounds memory. Overlapping predictions use separable squared-sine ramps at internal boundaries to avoid hard seams without attenuating true outer image edges. Eight-way dihedral self-ensemble is optional and must be reported as a compute-quality tradeoff. No reconstruction result is reported because no rights-reviewed source-disjoint high/low-resolution corpus is available.

6 Registered evaluation methodology

The strongest falsifier is a source-disjoint comparison against the best matched single family, concatenation plus MLP, uniform projected averaging, ordinary softmax gating, loss-free balancing, and host-risk-only/selective-prediction controls. Parameter-, activated-FLOP-, information-, and optimizer-budget matching are required. At least five frozen seeds are used unless a prospective power analysis demands more.

Primary outcomes are paired task quality and selective failure detection. Secondary outcomes are Brier score, ECE, risk-coverage, unsafe continuation, safe-case STOP, component-removal delta, OOD shift, corruption, missing modalities, runtime, peak memory, and energy proxy. Calibration thresholds are chosen only on calibration data. Protected test outcomes are inspected once. Failed runs, OOM, timeouts, NaNs, and resource rejections remain in the denominator. Confirmatory primary tests require multiplicity control and paired confidence intervals.

The repository's aligned dataset contract validates provenance, hashes, source identity, shape, finiteness, required fields, legal targets, and declared extensions. However, the actual aligned corpus does not yet exist. Therefore Checkpoint 7, confirmatory freeze, cannot be passed.

7 Bounded engineering case study

7.1 Data and frozen matrix

The only loaded corpus contains 36 proprietary smoke instruction records, with 12 each in train, validation, and test. All 12 capability categories occur once per split, and the manifest reports no exact or contiguous eight-token cross-split duplicate. The test split had been inspected before this study; all results are development-only.

The frozen matrix contained 24 CPU runs and 900 attempted epochs. Large models had 884,480 parameters; medium models 145,664; small models 35,552. We used byte tokenization, sequence length 192, batch size 4, gradient accumulation 2, AdamW at 10^{-3}, weight decay 0.01, gradient clipping at 1.0, and 30 attempted epochs. Five matched seeds compared cosine decay, constant rate, and final-epoch versus validation-best state. Three seeds tested packing and model scale. Learning-rate cosine annealing follows the broad schedule family introduced by SGDR [7], without warm restarts.

7.2 Results

Variant	Seeds	Parameters	Test loss, mean	Format rate, mean
Cosine large	5	884,480	3.13889	0.8333
Constant large	5	884,480	3.14858	0.8333
No restore large	5	884,480	3.16673	0.8333
No packing large	3	884,480	3.14122	0.8333
Cosine medium	3	145,664	3.44510	0.8333
Cosine small	3	35,552	4.61653	0.3889

Table 1: Exploratory results on an already observed 12-record smoke test. Exact match was 0 in all 24 runs.

Cosine minus constant test loss averaged -0.00969 across five paired seeds with a deterministic bootstrap 95% interval $[-0.01607, -0.00564]$. Cosine with validation-best restoration minus the same schedule at epoch 30 averaged -0.02784 with interval $[-0.03644, -0.02087]$. These intervals quantify seed variation in the frozen matrix; they do not repair test reuse or establish population-level coverage.

Loss fell with parameter scale, but the comparison was not compute matched. Peak process RSS was 294,780,928 bytes in the shared process for every recorded evaluation, so it is not sensitive enough to compare variants. Wall time was contaminated by concurrent host load and is reported only in raw artifacts.

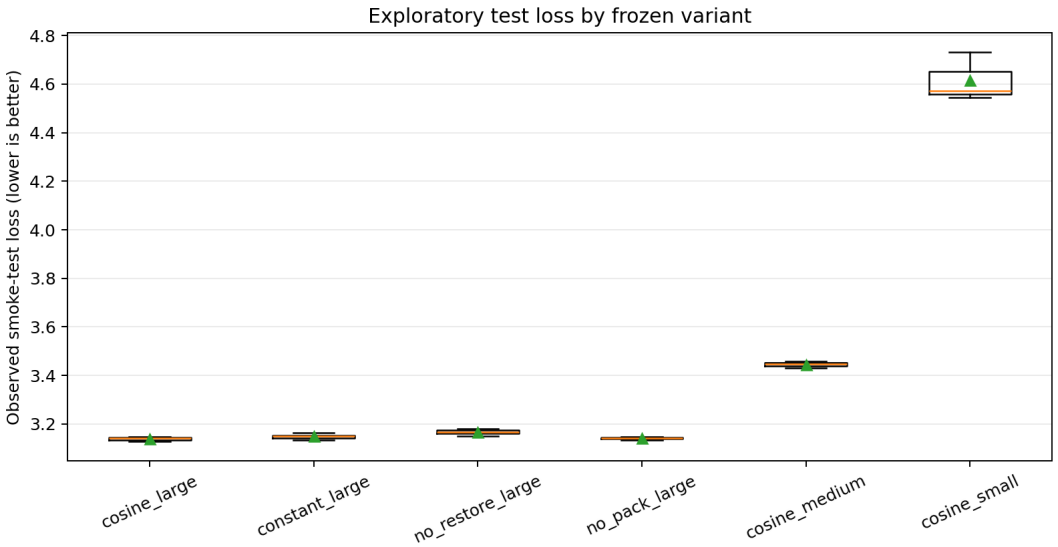


Figure 2: Per-seed exploratory smoke-test loss. The study is not confirmatory.

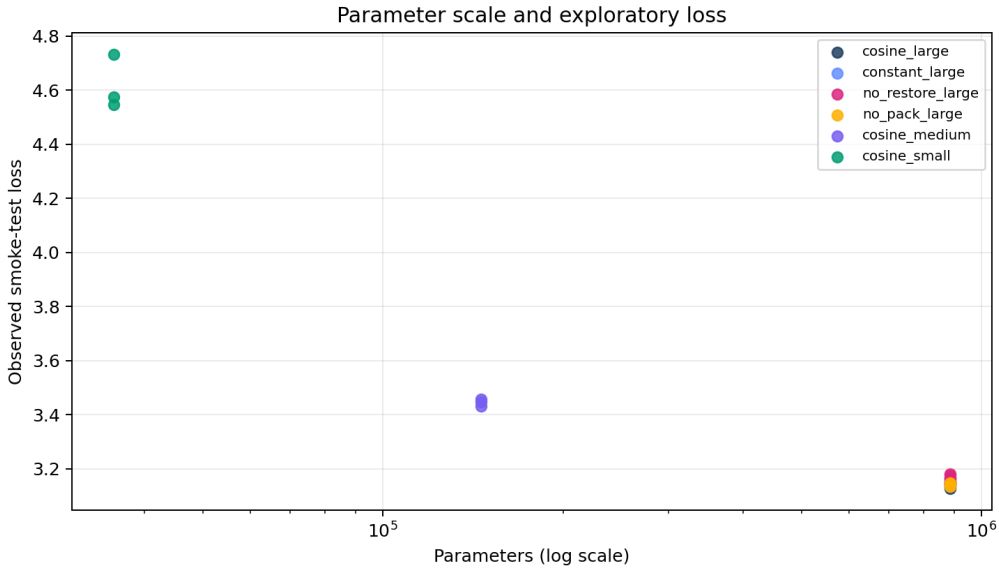


Figure 3: Parameter scale versus observed test loss. This is a confounded scale ablation, not a scaling law.

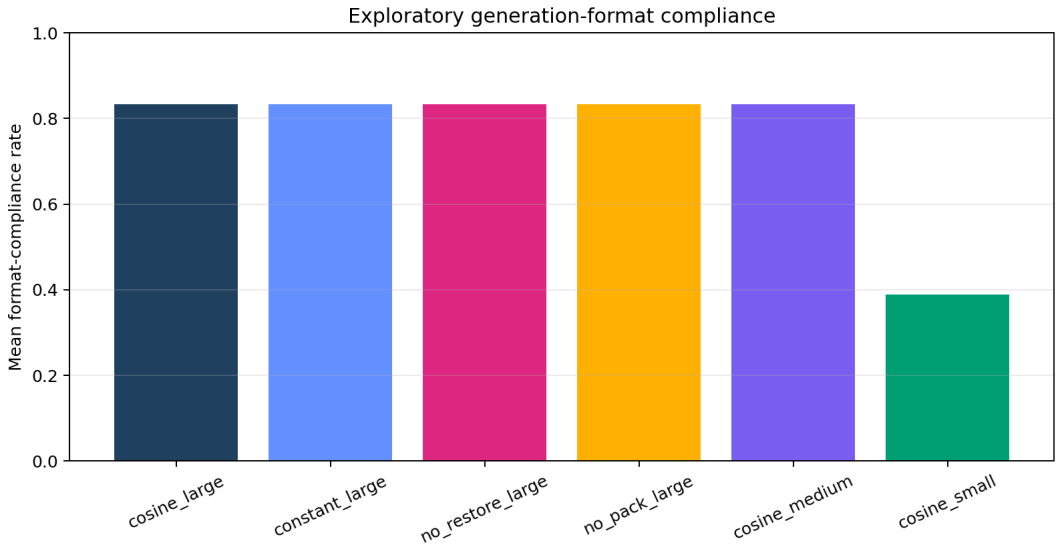


Figure 4: Generation-format compliance was stable for medium and large models but highly variable and lower for the smallest model.

8 Failure and adversarial analysis

The first matrix attempt was blocked at 91.2% swap utilization against a frozen 90% ceiling. All 24 rejections were retained. The analysis then failed on an empty completed set; that attempt is invalid, the defect is documented, and the corrected run used a new artifact root without changing the guard.

The packing intervention was inactive: packing-on and packing-off logs were bit-identical for shared seeds because examples were already too long to combine under the 192-token cap. This ablation is non-informative. Most importantly, exact match remained 0% for all runs. Lower cross-entropy and acceptable formatting do not imply useful instruction capability.

Conceptual threats remain substantial: the fusion mechanism may not be novel as a whole; component influence is not identifiable under substitutable representations; added parameters, compute, and information can explain

gains; joint losses can create gradient conflict; risk labels can be ambiguous; calibration can fail under shift; and a simpler linear or uniform fusion may be sufficient. The JEPA objective may also be unnecessarily complex. Each threat has a registered control in the protocol, but none has valid outcome data yet.

9 Software verification

The final repository verification before paper generation passed 181 tests with 86.08% branch coverage against an 85% floor. Ruff and strict mypy passed. The frontend test and production build passed, dependency audits reported no known vulnerabilities in audited packages, and wheel/sdist construction plus metadata checks passed. Tests cover shapes, gradients, finite losses, forbidden transitions, monotone risk, STOP behavior, extension channels, content-addressed data, immutable checkpoints, JEPA tiling, and end-to-end reference execution. These are software claims, not scientific validation.

10 Limitations and disposition

No aligned multimodal/temporal/governance dataset exists in the repository. The smoke test is tiny and contaminated for fresh confirmation. No OOD, intervention, corruption, or source-independent JEPA evaluation was possible. The literature search was broad but not independently reproduced. Reference-scale behavior may not survive scaling.

Accordingly, the central Olympus result is **INCONCLUSIVE / NOT EVALUATED**. The bounded engineering case study is **MIXED**: two training safeguards improved an already observed loss measure, scale correlated with lower loss, packing was non-informative, and exact-match capability remained zero. Promotion, deployment, and confirmatory claims are not authorized.

11 Reproducibility and release statement

The canonical state records the question, budget, protected boundary, data hashes, failures, decisions, and history. The frozen matrix manifest precedes execution. Each run retains a configuration hash, checkpoint hash, metrics, evaluation, elapsed time, and memory snapshot. Aggregate JSON, CSV, figures, failure log, and completion receipt are content-addressed. Reproduction commands are provided with the source. This manuscript and package are arXiv-ready as a registered architecture and preliminary engineering report, not as a positive experimental paper. External submission was not performed.

12 Conclusion

Olympus makes a narrow architectural promise: learned evidence cannot lower host risk, legal transitions remain host-owned, and every component-removal decision is auditable. The implementation and engineering pipeline work end to end under bounded conditions. The central empirical question remains unanswered until a rights-reviewed aligned corpus and untouched protected test exist. Preserving that boundary is the principal scientific result of the current project state.

A Registered confirmatory matrix

The future study must include best-family, concatenation, uniform, ordinary-gate, loss-free balance, host-only risk, entropy, SelectiveNet, deep-ensemble, and every leave-one-family-out baseline. It must test modality deletion, corruption, source and temporal shift, risk-label perturbation, gate temperature, objective weights, dataset and model scale, frozen versus joint backbones, and JEPA loss terms. All configurations, practical effect thresholds, exclusions, and stopping rules are frozen before protected outcomes.

B Artifact inventory

Primary artifacts are the canonical JSON state, protocol, literature review, result audit, evidence map, reproduction instructions, immutable v1 failure attempt, corrected 24-run root, model checkpoints, aggregate statistics,

figures, source, bibliography, PDF, and arXiv package. Proprietary dataset records are not included in a public submission unless distribution rights are established.

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